

SUMIT YADAV

+44 7823 712507 | sumityadav8976502030@gmail.com | linkedin.com/in/yadavsumit2003 |
github.com/SumitYadav2003 | Leicester, UK

PROFESSIONAL SUMMARY

MSc Advanced Computer Science student (University of Leicester) and full-stack developer — Django/PostgreSQL, AWS, CI/CD — who takes projects all the way to live production.

- **29 functional and 10 non-functional requirements met**, verified through 64 structured test scenarios at a 100% pass rate, on CulinaAI — an AI-integrated Django/PostgreSQL recipe platform.
- **Independently deployed CulinaAI to live production** — real cloud infrastructure experience most junior candidates only read about in a job description.
- **Independently designed and built CarVerse**, a full PHP/MySQL car dealership platform (BSc final year project) spanning authentication, admin CRUD, search, comparison and booking features.

KEY SKILLS

Languages & Frameworks: Python, Django, JavaScript, HTML, CSS, Bootstrap

Engineering Fundamentals: Object-Oriented Programming, System Design

Databases: PostgreSQL, MySQL, SQL, relational schema design

Cloud & DevOps: AWS (S3, IAM), Render, GitHub Actions CI/CD, Docker, Git, GitHub, GitLab

APIs & Integration: RESTful API design and integration (OpenAI API)

Testing & Quality: Automated unit testing (Django TestCase), scenario-based functional/integration/validation testing, defect tracking and resolution

Security & Architecture: Role-based access control (RBAC), authentication, requirements engineering, iterative/agile development

PROJECT EXPERIENCE

CulinaAI — AI-Assisted Recipe Generation and Meal Personalisation System

Jun 2026 – Sep 2026

MSc Individual Dissertation, University of Leicester | Django, PostgreSQL, JavaScript, Bootstrap, OpenAI API, AWS S3, Git

Live Demo: culinaai.onrender.com | Code: github.com/SumitYadav2003/culinaai

- **Met all 29 functional and 10 non-functional requirements** by architecting a full-stack Django/PostgreSQL platform integrated with the OpenAI API across seven personalisation parameters, from dietary allergies to available cooking equipment.
- **Cut unsafe AI output to zero** across every validation test scenario by designing an 8-point automated validation engine that screens each generated recipe for allergy safety, dietary compliance, structure and equipment constraints before it reaches storage.
- **Passed 64/64 structured test scenarios** across 6 categories (functional, integration, AI-validation, pantry/shopping, modification, interface) by designing and executing a full functional, integration and validation test programme.
- **Shipped 4 additional production modules** — Smart Pantry with expiry tracking, refrigerator-scan detection, side-by-side recipe revalidation, and a real-time AI cooking assistant — as independently testable Django apps sharing one recipe/pantry data model.
- **Deployed the system to live production** on Render with a GitHub Actions CI/CD pipeline and persistent AWS S3 image storage under a least-privilege IAM policy — turning an academic prototype into a real, publicly demoable service.

CarVerse — PHP/MySQL Car Marketplace and Dealership Website

2023 – 2024

BSc Final Year Project, PTV's Mulund College of Commerce | PHP, MySQL, HTML/CSS, JavaScript, XAMPP

Code: github.com/SumitYadav2003/carverse

- **Built a full car dealership website end to end** — an admin-manageable car catalogue (add/edit/delete listings with multi-image upload) backed by MySQL, plus a 40+ page vehicle showcase spanning multiple categories.
- **Implemented dual-role session authentication**, separating public-user login from an admin panel that gates all car-inventory management.
- **Delivered supporting customer features** — MySQL-backed search, side-by-side vehicle comparison, test-drive/service appointment booking, and a reviews/contact-message system.

EDUCATION

MSc Advanced Computer Science

Sep 2025 – Sep 2026

University of Leicester, UK

BSc Computer Science

2021 – 2024

PTV's Mulund College of Commerce (Autonomous), India | CGPA: 8.53/10